Coulomb excitation 1969Ha17,1977Sc36

1969Ha17: 51-61-MeV $^{35}Cl^{5,6+}$ beams of 80 nA were produced from the Chalk River MP tandem accelerator and bombarded 9.6-mg/cm^2 carbon and 5-mg/cm^2 ^{24}Mg. De-excitation γ rays were detected using two 40-cm^3 Ge(Li) detectors at 125° and at 0° or 90°. Measured Eγ, Iγ, $\gamma(\theta)$ anisotropy, γ-ray yields. Deduced levels, mixing ratio, transition strengths, $T_{1/2}$ for 2 levels using the Doppler Shift Attenuation Method (DSAM).

1977Sc36: 53-56-MeV $^{35}Cl^{6+}$ beams of 60 nA were produced from a single MP Tandem Van de Graaff tandem accelerator at the three-stage facility of Brookhaven National Laboratory. Targets were 23Na and 27Al. De-excitation γ rays were detected using a 55$^-$ cm^3 coaxial Ge(Li) detector at 0°. Measured Eγ, I$\gamma(\theta)$, γ-ray yields. Deduced levels, mixing ratios, transition strengths, $T_{1/2}$ for the levels of 1220, 1762 and 2650 keV using the Doppler Shift Attenuation Method (DSAM).

^{35}Cl Levels

Values of B(E2)↑ from 1969Ha17 and 1977Sc36 are from absolute cross-section measurements.

E(level)†	J$^\pi$†	$T_{1/2}$	Comments
0	3/2$^+$		
1219	1/2$^+$	0.150 ps *20*	B(E2)↑=0.00076 *7* $T_{1/2}$: Lifetime τ=0.216 ps *28* from weighted average of τ=0.16 ps *5* (1969Ha17, DSAM) and τ=0.230 ps *25* (1977Sc36, DSAM). B(E2)↑: weighted average of 0.00081 *9* (1969Ha17) and 0.00073 *7* (1977Sc36).
1763	5/2$^+$	0.42 ps *4*	B(E2)↑=0.0106 *7* $T_{1/2}$: Lifetime τ=0.61 ps *5* from weighted average of τ=0.57 ps *7* (1969Ha17, DSAM) and τ=0.63 ps *5* (1977Sc36, DSAM). B(E2)↑: weighted average of 0.0117 *9* (1969Ha17) and 0.0102 *5* (1977Sc36).
2646	7/2$^+$	0.18 ps *6*	B(E2)↑=0.0029 *7* (1977Sc36) $T_{1/2}$: Lifetime τ=0.26 ps *9* (1977Sc36, DSAM).

† From the Adopted Levels. Energies are rounded values.

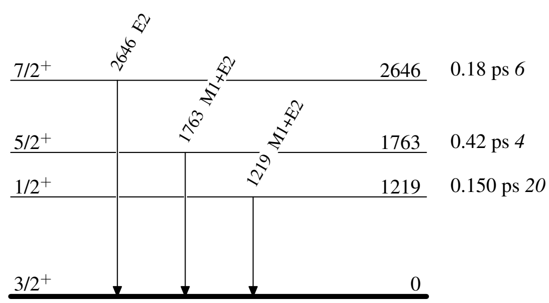
$\gamma(^{35}$Cl)

E$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.†	δ	Comments
1219	1219	1/2$^+$	0	3/2$^+$	M1+E2	0.093 *10*	δ: deduced by the evaluators from adopted B(E2)↑=0.00076 *7* in this dataset and $T_{1/2}$=0.119 ps *14* from Adopted Levels. Other: 0.106 *7* deduced by 1977Sc36 based on their measured B(E2)↑ and lifetime.
1763	1763	5/2$^+$	0	3/2$^+$	M1+E2	−2.95 *45*	δ: from $\gamma(\theta)$ anisotropy in 1969Ha17. Other: 1.8 +*10*−*4*; deduced by the evaluators from adopted B(E2)↑=0.0106 *7* in this dataset and $T_{1/2}$=0.36 ps *3* from Adopted Levels; −2.7 *7* from 1977Sc36 deduced from their measured B(E2)↑ and lifetime. δ: from adopted B(M1) and adopted lifetime in this dataset.
2646	2646	7/2$^+$	0	3/2$^+$	E2		

† From Adopted Gammas. Energies are rounded values.

Coulomb excitation 1969Ha17,1977Sc36

Level Scheme



$^{35}_{17}Cl_{18}$